GRADE 11 / Life Sciences / English

Photosynthesis: test an explanation

01 Understand

Learning goals: State the inputs and products • Interpret a limiting factor • Plan a controlled investigation

Photosynthesis uses light energy to make glucose from carbon dioxide and water, releasing oxygen. Chlorophyll absorbs light in chloroplasts.

A limiting factor restricts the rate even when other requirements are available. Increasing light may eventually stop increasing the rate if another factor becomes limiting.

A fair investigation changes one independent variable, measures a dependent variable and controls other relevant conditions. Repeat measurements to improve reliability.

Illustrative pondweed results

Lamp distance (cm)	Bubbles per minute	Interpretation
10	24	higher light
20	15	lower light
30	7	lowest light

02 Follow the reasoning

Interpret the results and identify one weakness of counting bubbles.

1. As lamp distance increases, bubble count falls, suggesting reduced photosynthesis as light decreases.

2. Distance is changed; bubble count is measured. Keep plant size, water temperature and carbon dioxide availability controlled.

3. Bubble sizes vary, so counts are an imperfect estimate of oxygen volume. Measuring gas volume over time gives a stronger comparison.

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Practise with purpose

03 Try before checking

Use paper or your notebook. Show your reasoning. For written responses, use evidence and explain your choices. Suggested time: 15–20 minutes.

1. Name two raw materials of photosynthesis.

2. Why might a lamp heat confound the investigation?

3. Why does a rate plateau not mean that photosynthesis has stopped?

04 Check and correct

1. Carbon dioxide and water.

2. Temperature also affects enzyme-controlled reactions, so the result may reflect two changing variables.

3. The rate can remain positive but constant because another factor limits further increase.

Common mistake to avoid

Glucose is a product; sunlight supplies energy and is not a raw material with mass in the equation.

Reflect and revisit

Identify one correction in your own words. Attempt a similar question tomorrow without looking at the guide. If you are still unsure, take your attempt to a teacher or tutor.

Coverage and sources

CAPS topic focus: Life processes in plants and animals: photosynthesis

DBE CAPS, Grades 10–12: Life Sciences. Section 3 content and skills. This is a focused supplement, not a complete syllabus or prescribed literature replacement.

Open the official DBE curriculum reference

Content and visuals: original SmartBridge FET. AI-assisted authoring; subject-teacher and language-specialist review pending. Report an issue with the pack ID lisc-11.